

IoT & Sensor Data Management

Ambient Intelligence and Domotics

Master Degree: Artificial Intelligence for Science and Technology

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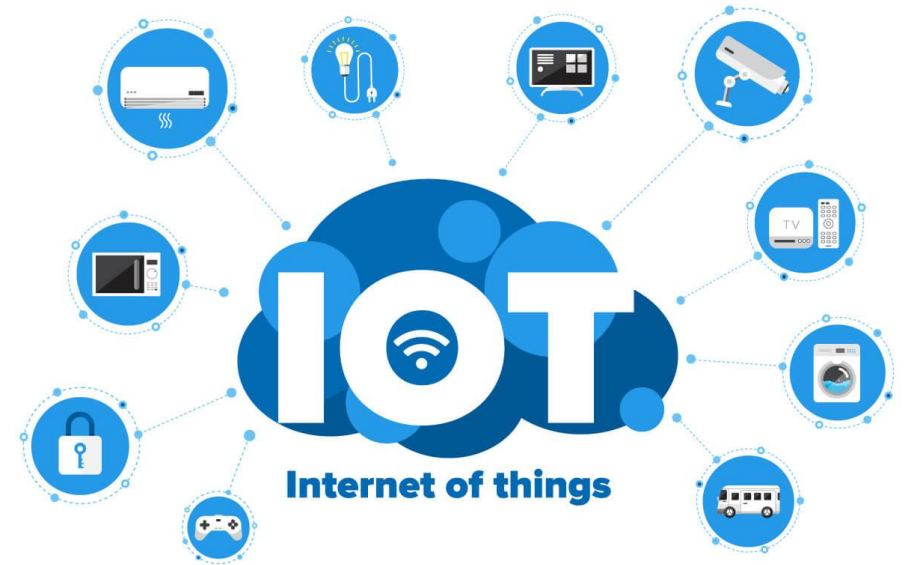
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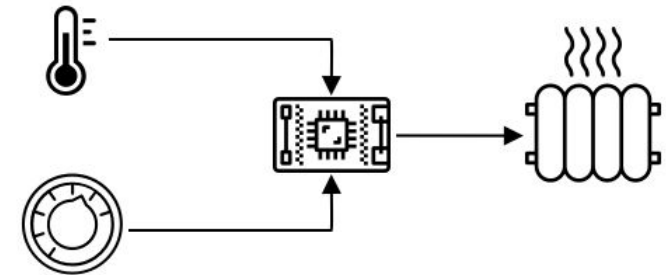
Internet of Things (IoT)

A set of physical “smart” objects (*things*) equipped with sensors, software, and other technologies, collecting data from the environment and exchanging it over a computer network to achieve a specific goal.



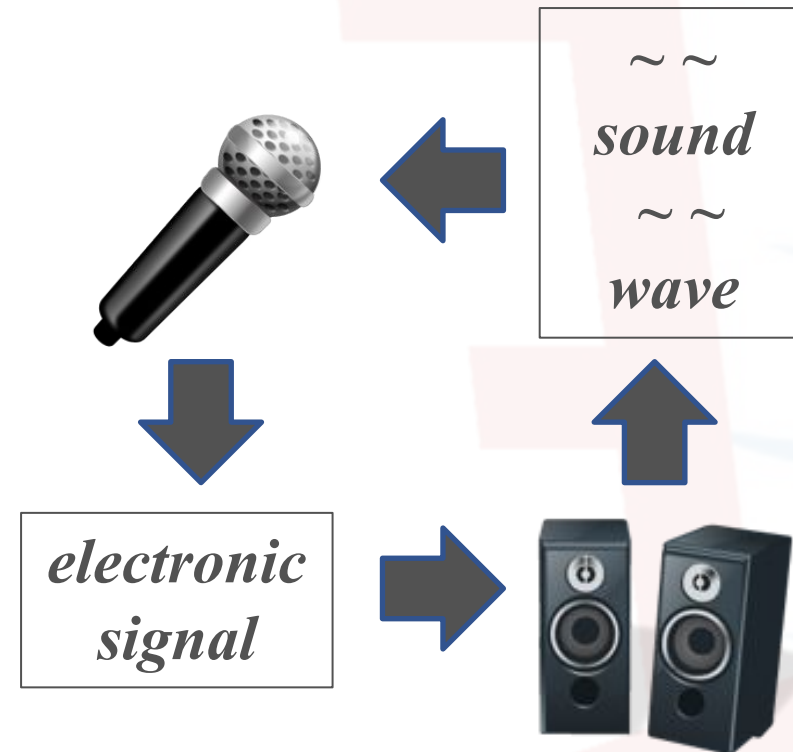
The “Thing”

- A device that can interact with the physical world (sensing or actuating)
- Things can be in different forms, from dedicated hardware sensing a specific phenomena, to general purpose devices (e.g., smartphone)



Transducer

- Device that transforms one form of energy into another
- Input transducers:
sensors
- Output transducers:
actuators

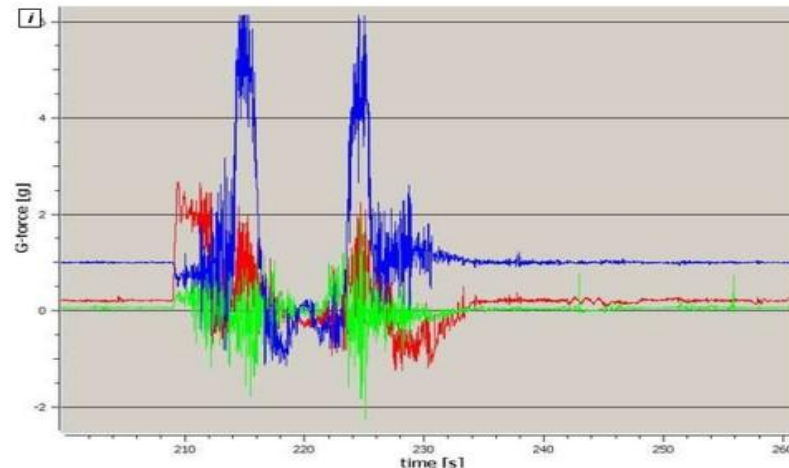


How sensors work

- Sensing process



- Sensors detect physical changes and produce an electrical signal
- Output: data stream
 - Example: Accelerometer on a smartphone with readings at 100Hz

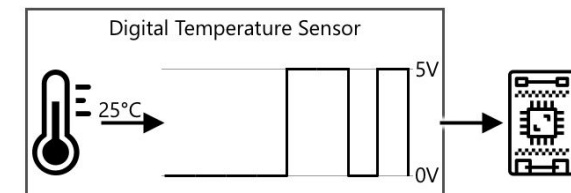
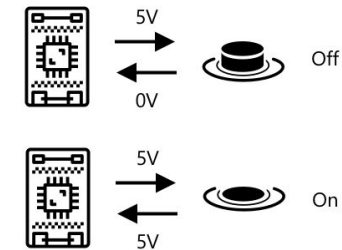
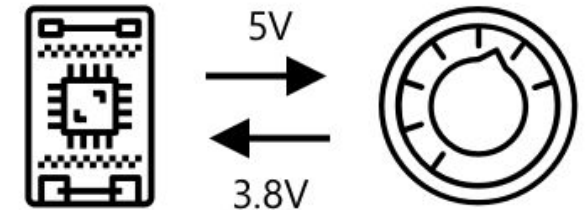


Sensor types

Analog sensors: they receive voltage from the IoT device and return an adjusted voltage value based on the sensed phenomena. The returned voltage is the analog representation of the sensed measurement.

- Analog signals need to be digitalized (IoT devices can only understand 0s and 1s).
- This is done with specific components called ADC (analog-to-digital converters)

Digital sensors: They output a digital signal, either by only measuring two states (ON and OFF) or by using a built-in ADC.



Actuators

- Actuators are also a particular type of transducers
- They convert energy into an action (motion, switch), ... upon receiving a command from a control system
- Examples: LED, speakers, relay, screens, stepper motors

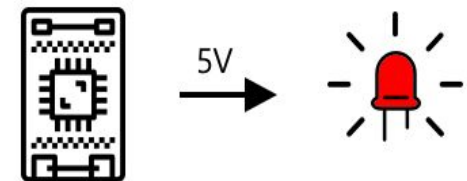
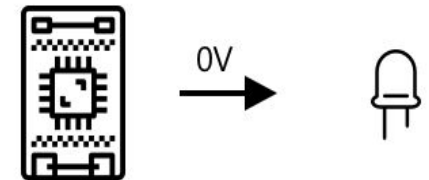
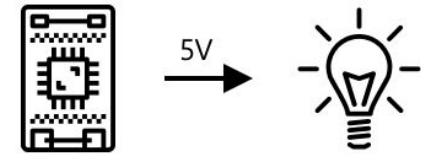
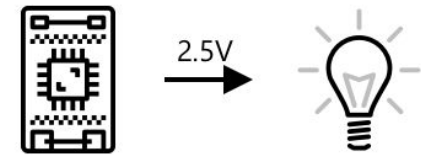


Actuators Types

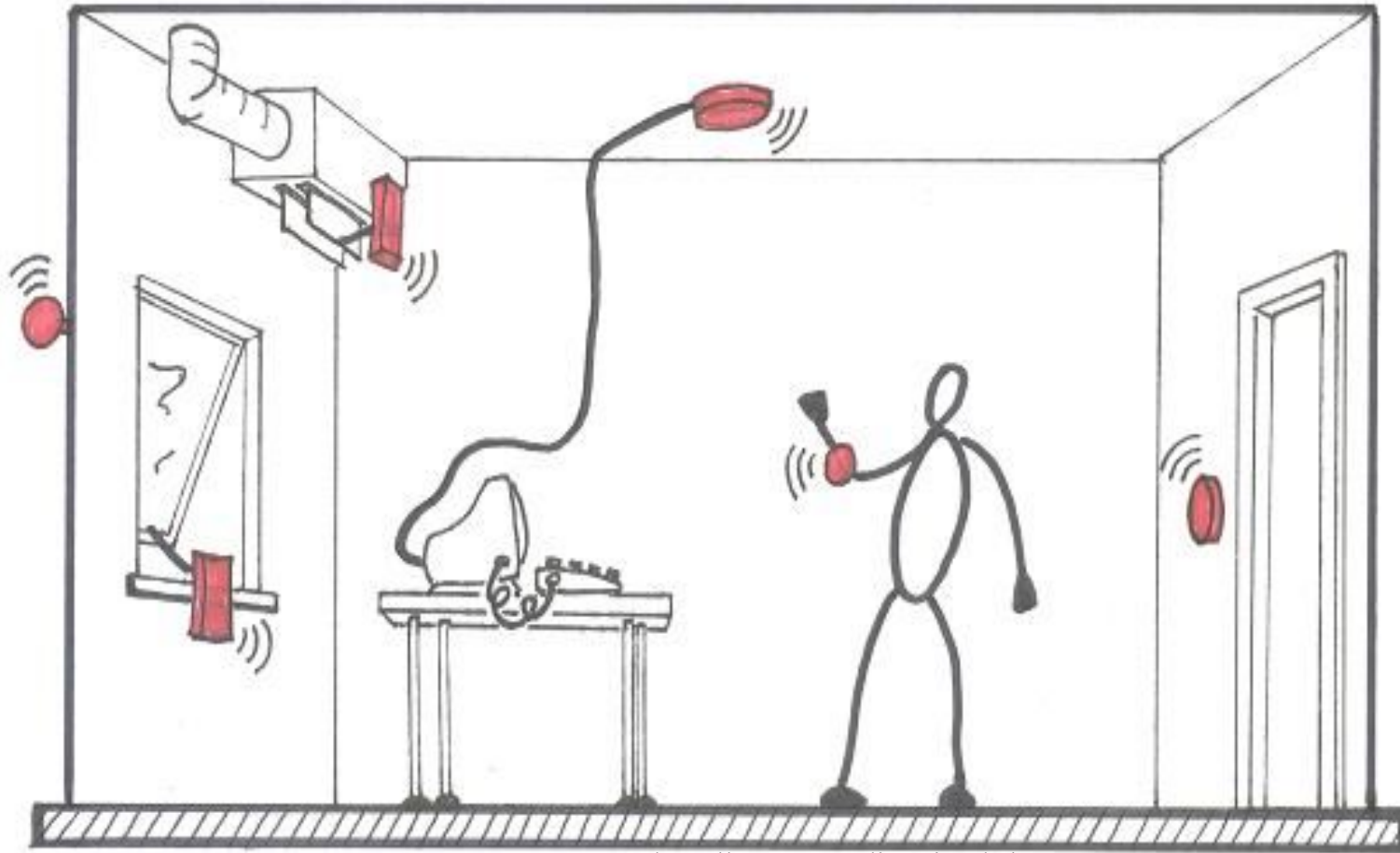
Analog actuators: take an analog signal and convert it into some kind of interaction, where the type of interaction changes based on the voltage supplied.

- A DAC (digital-to-analog) component is required to make devices interact with actuators.
- They convert 0s and 1s into a voltage.

Digital actuators: They either have two states controlled by a high or low voltage or have a built-in DAC to convert a digital signal to an analog one.



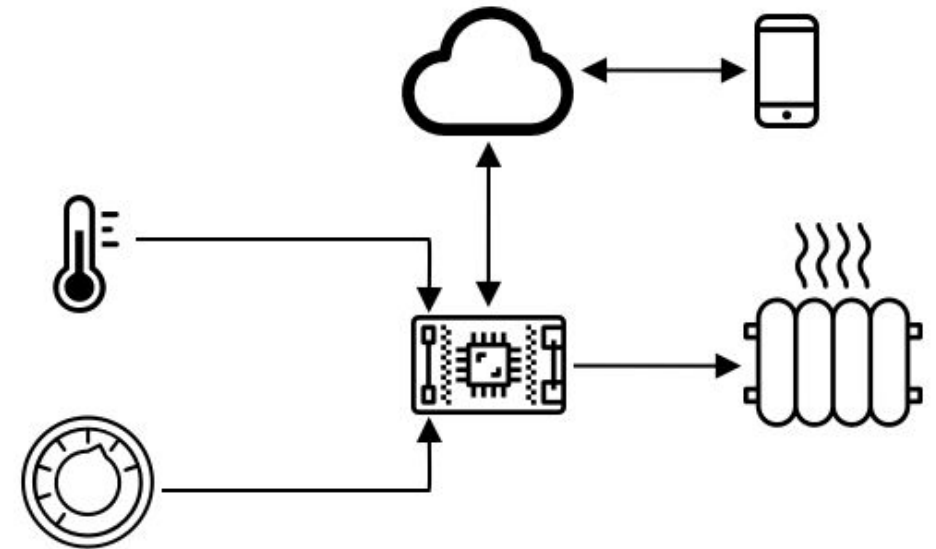
Sensors and actuators



Source: <http://resenv.media.mit.edu/>

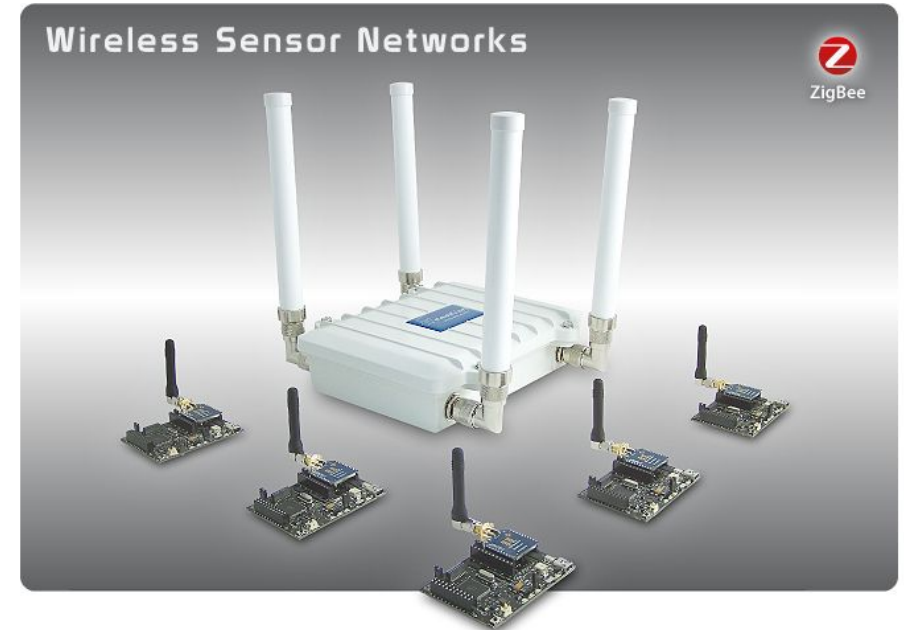
The “Internet”

- The **Internet** part of an IoT system consists of one or more applications:
 - sending/receiving data to/from IoT devices
 - processing data from IoT devices (e.g., using AI methods) to perform some analysis and perform decisions
- IoT devices usually don't communicate directly with the cloud, but they communicate through a local **gateway**

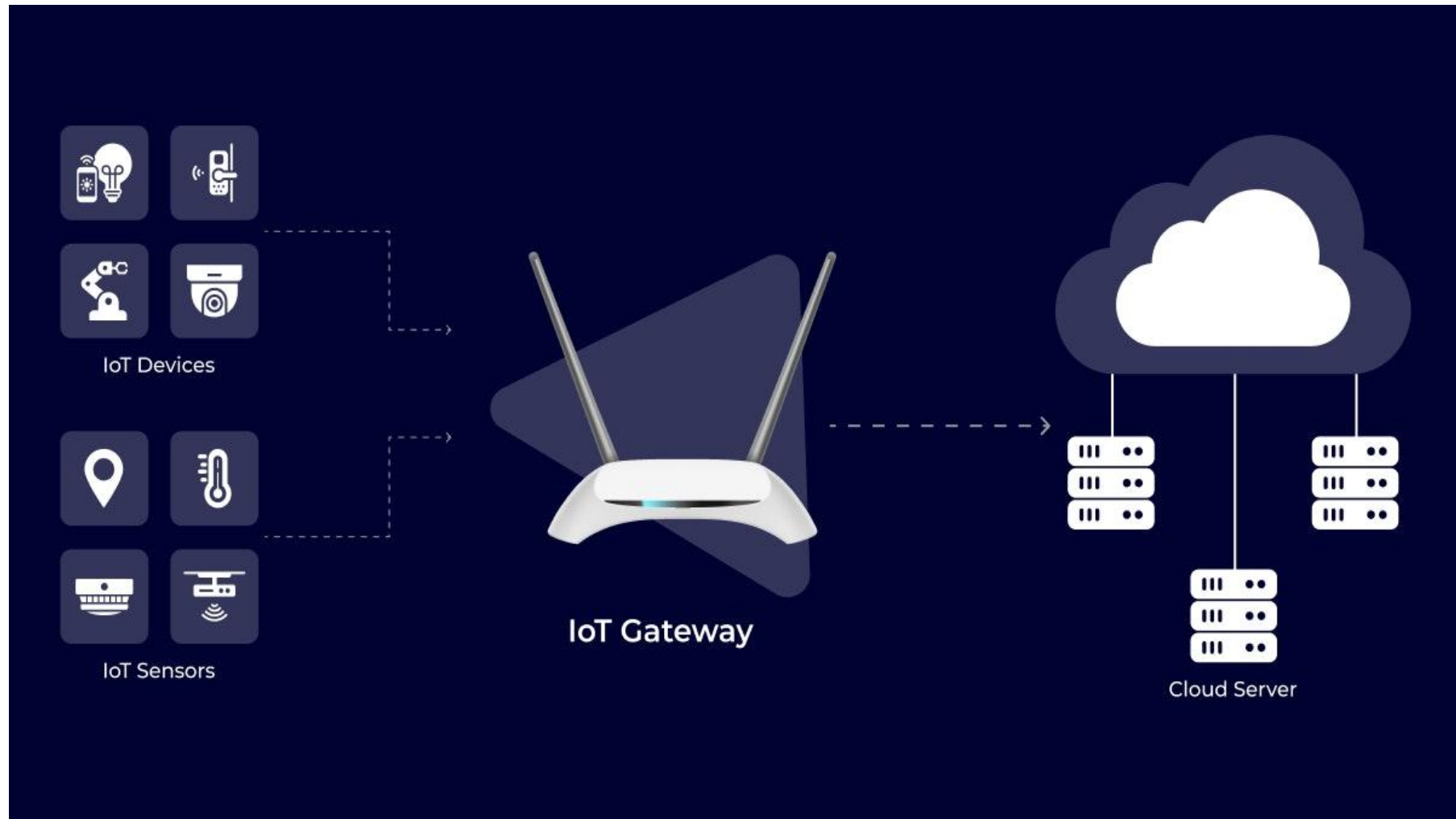


IoT Gateway

- Role: collecting data from physically close IoT devices
- It may connect to other networks (e.g., internet, other gateways)
- Possibly with embedded DB and local data processing
- Transmit data to processing units (cloud/server)

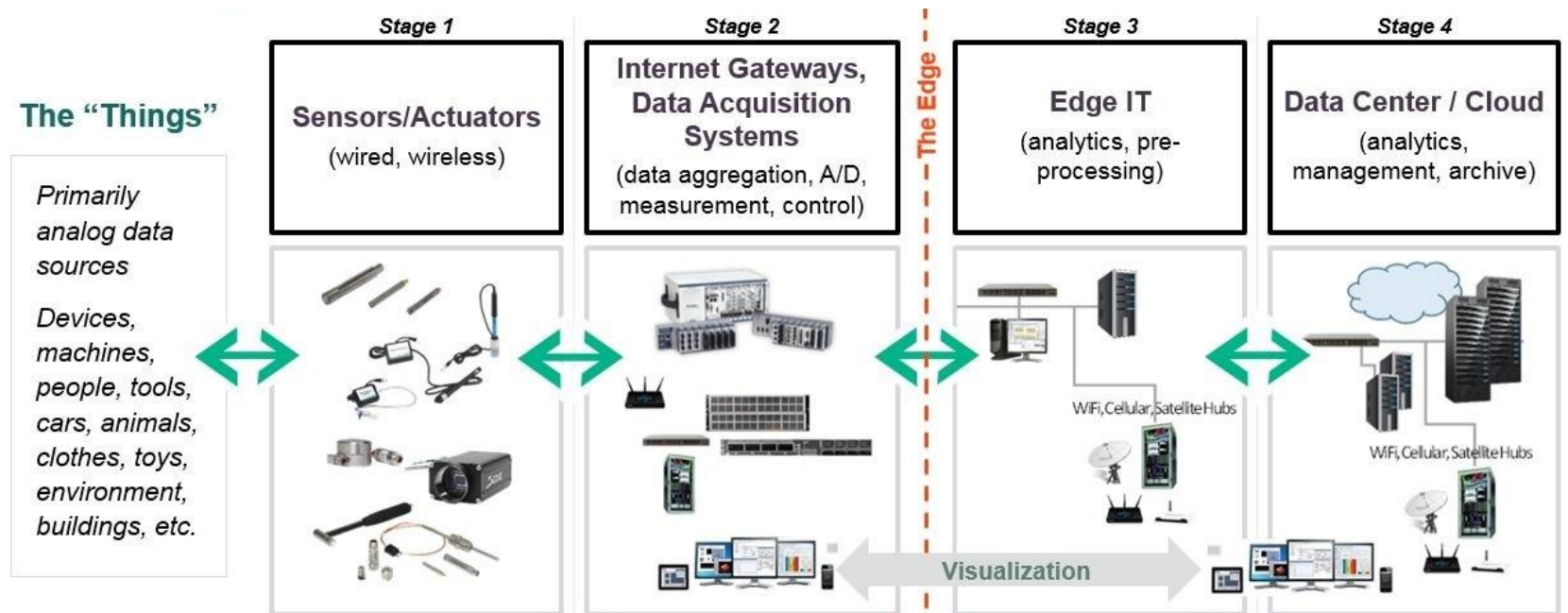


Three-stage IoT architecture



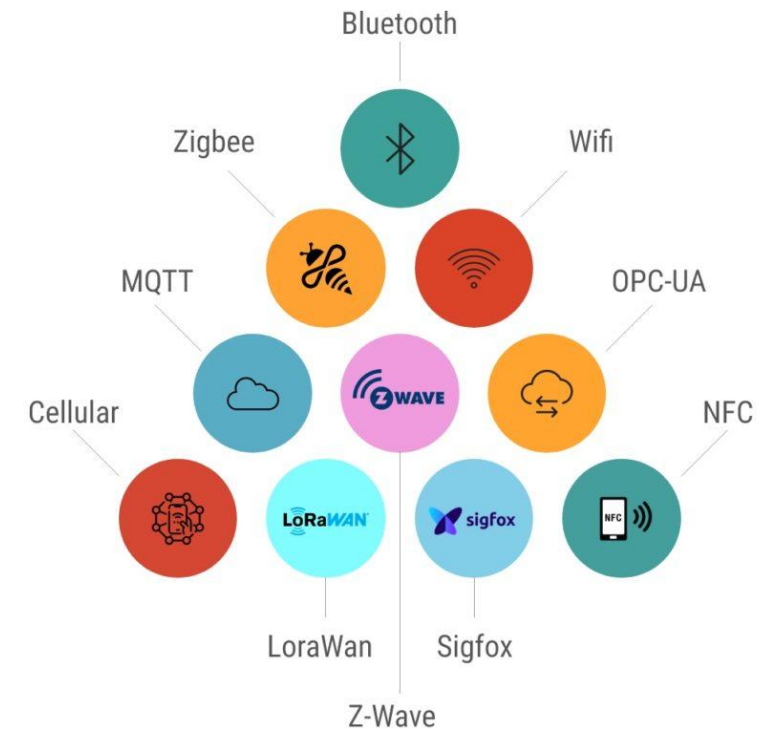
Four stage IoT architecture

- Edge nodes are machines close to the IoT network
 - e.g., the gateway covers the devices in a room, while there is an Edge device for the whole building
- They have more computational capabilities than the gateway and they can be used to:
 - reduce latency
 - pre-processing a smaller set of data before sending them to the cloud

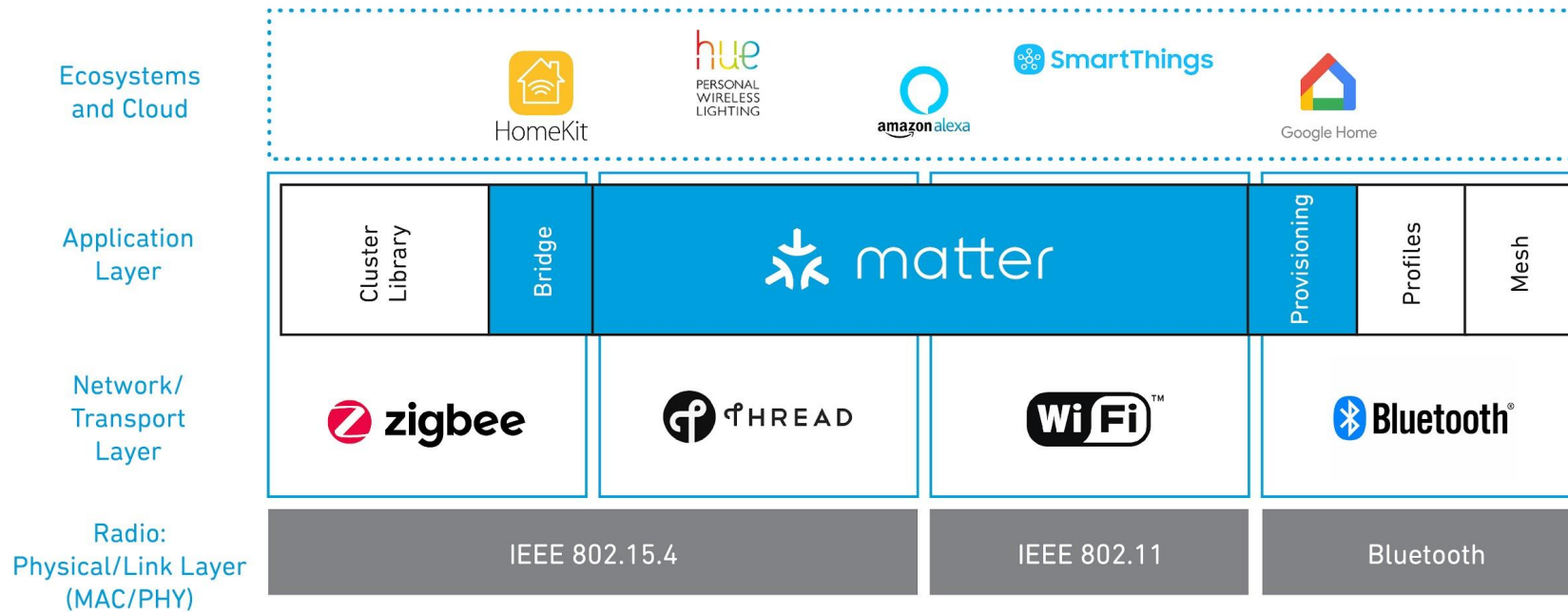


Communication

- Each IoT device may use very different networking interfaces to communicate with the gateway
- Each network modality has its own peculiarities
- Integration is challenging!



Matter: how to make IoT devices “talk” between them

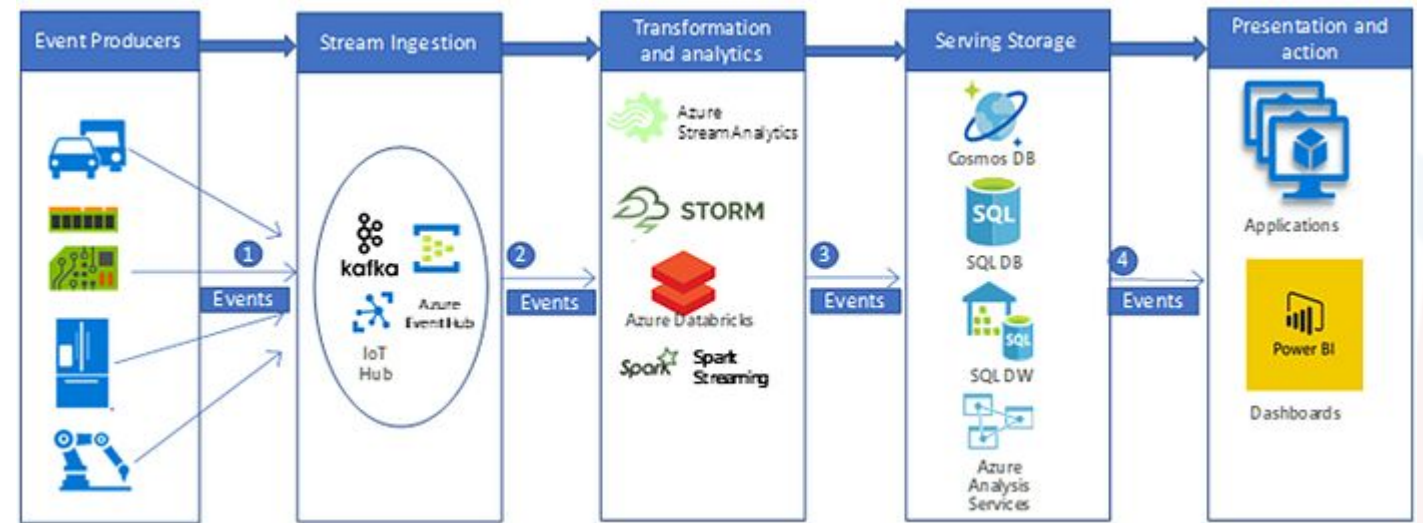


Types of IoT

- **Consumer IoT:** to be used for personal uses (e.g., at home).
 - e.g., smart speakers, smart heating systems, robotic vacuum cleaners, smart appliances
- **Commercial IoT:** devices used in the workplaces
 - e.g., occupancy sensors and motion sensors for smart energy management, temperature sensors for cold storages, localizing vehicles for transportation
- **Industrial IoT:** devices to control machinery on large scale and for predictive maintenance
 - e.g., temperature, vibration, rotation speed
- **Infrastructure IoT:** monitoring local and global infrastructures like smart cities or smart grids
 - e.g, pollution sensors, traffic monitoring sensors, energy consumption

IoT is also Big Data Analysis

- IoT devices may generate a huge amount of sensor data!
- Hence, Big Data Analysis pipelines are usually needed



Brief History of IoT

1982: the first IoT device

- A coca-cola vending machine connected to the internet
- It was possible to connect through the network (ARPANET)
 - to check if there was a drink available and if it was cold

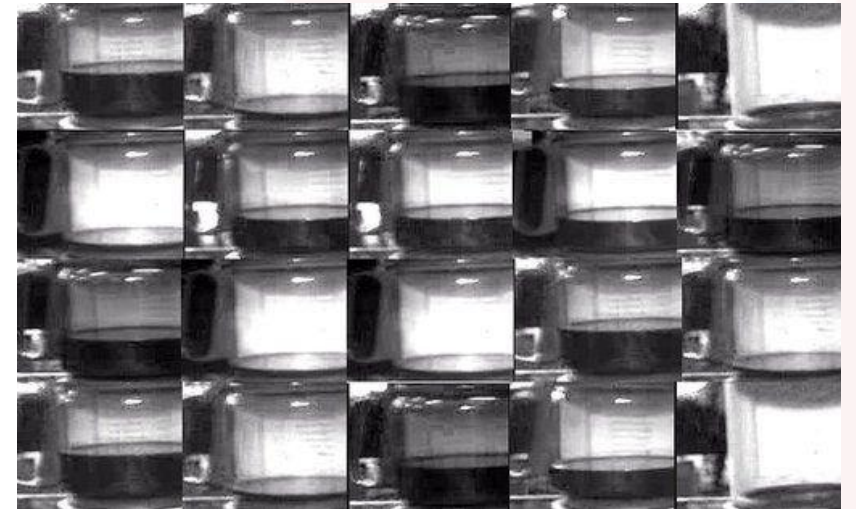


1990: a toaster is the first device connected to Internet



1993: The first webcam

- First webcam in the web to capture a coffee pot the Cambridge University
- It was stationed in a specific room and many of the researchers worked in different labs and on different floors
 - The problem: *"They would often turn up to get some coffee from the pot, only to find it had all been drunk"*



1998: The IoT fountain

- Mark Weiser (father of ambient intelligence) created an IoT fountain out of his office
 - The flow and the height of the water mimicked the price trends and volume of the stock market in real-time!



1999: the term “IoT” is coined

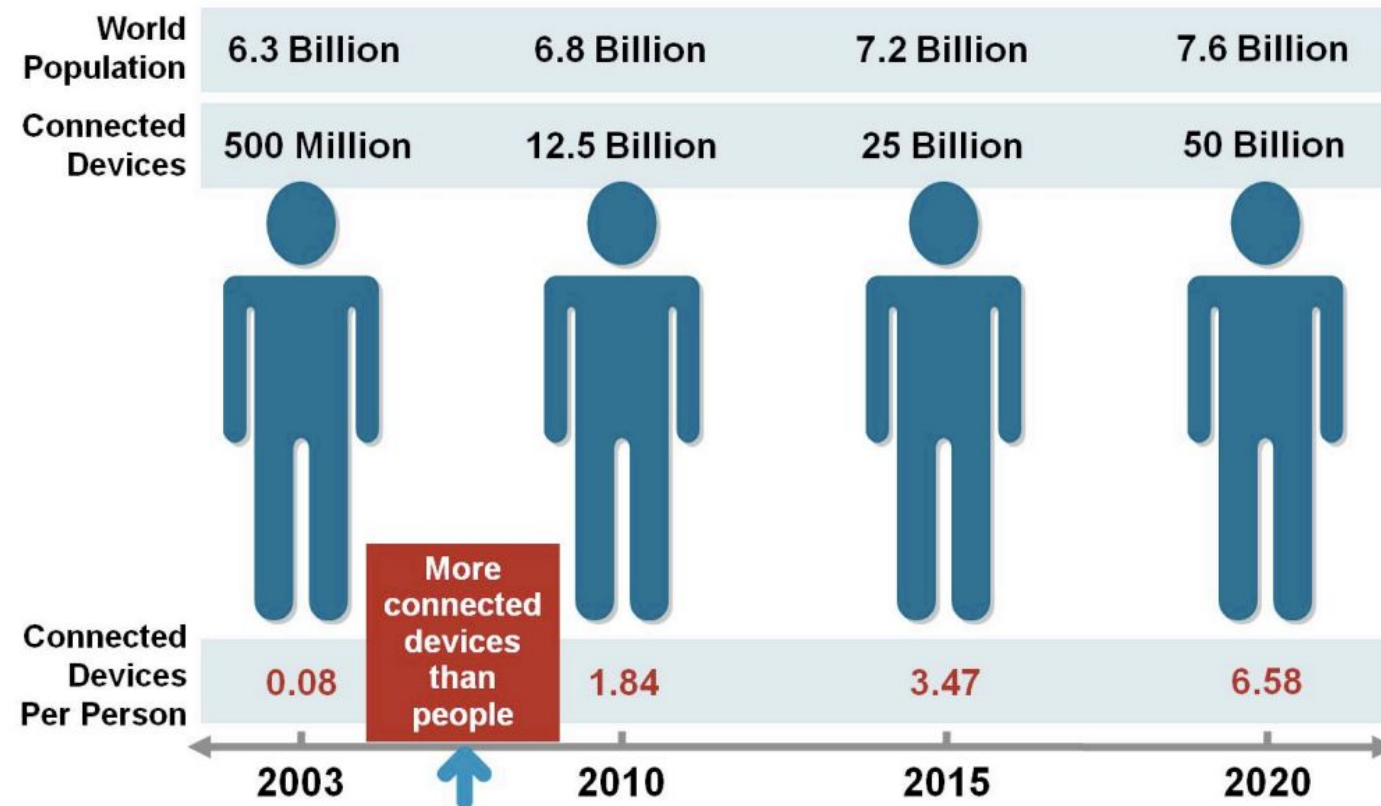
1999!

**“THE INTERNET OF THINGS IS
ABOUT EMPOWERING COMPUTERS
...SO THEY CAN SEE, HEAR
AND SMELL THE WORLD FOR
THEMSELVES”**

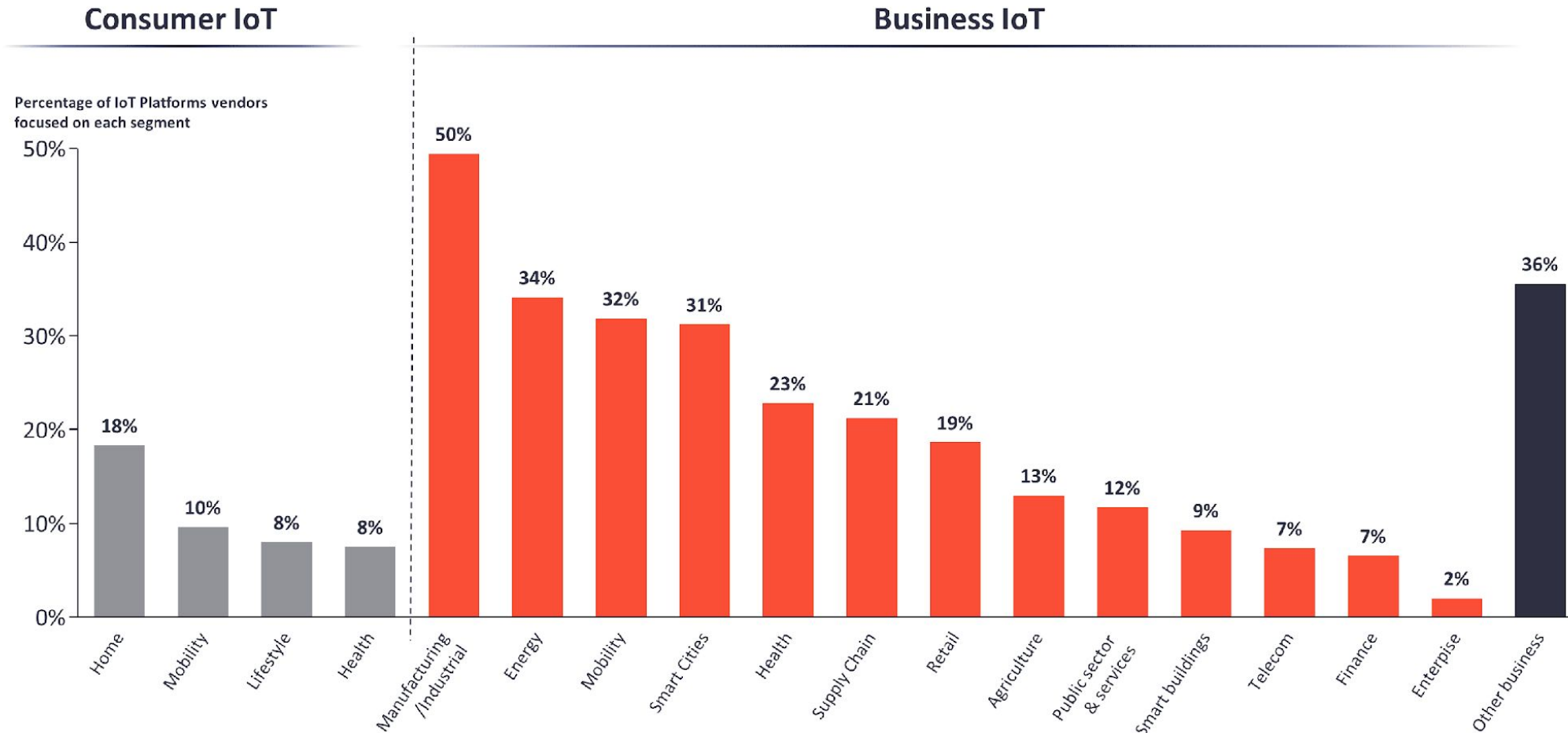
**KEVIN ASHTON
INVENTOR OF THE TERM
“INTERNET OF THINGS”**



From 2000s: Things > Humans



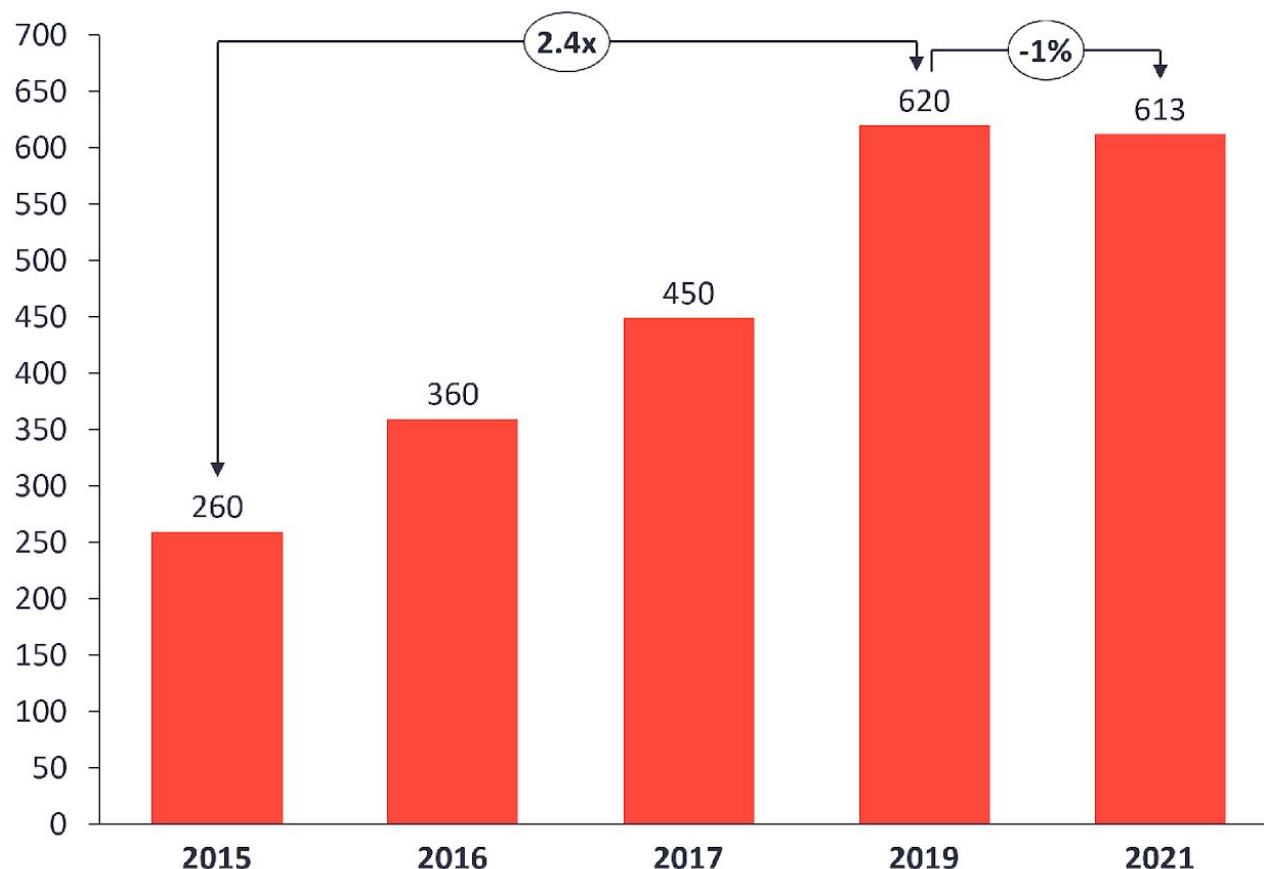
Number of Identified IoT Platforms – By industry (Dec 2019)



Note: Percentages do not add up to 100% as most companies focus on several segments **Source:** IoT Analytics 2020 List of IoT Platform Companies, n=620

Number of publicly known "IoT Platforms" (2015-2021)

Number of publicly known "IoT Platforms" (IoT Analytics Research)

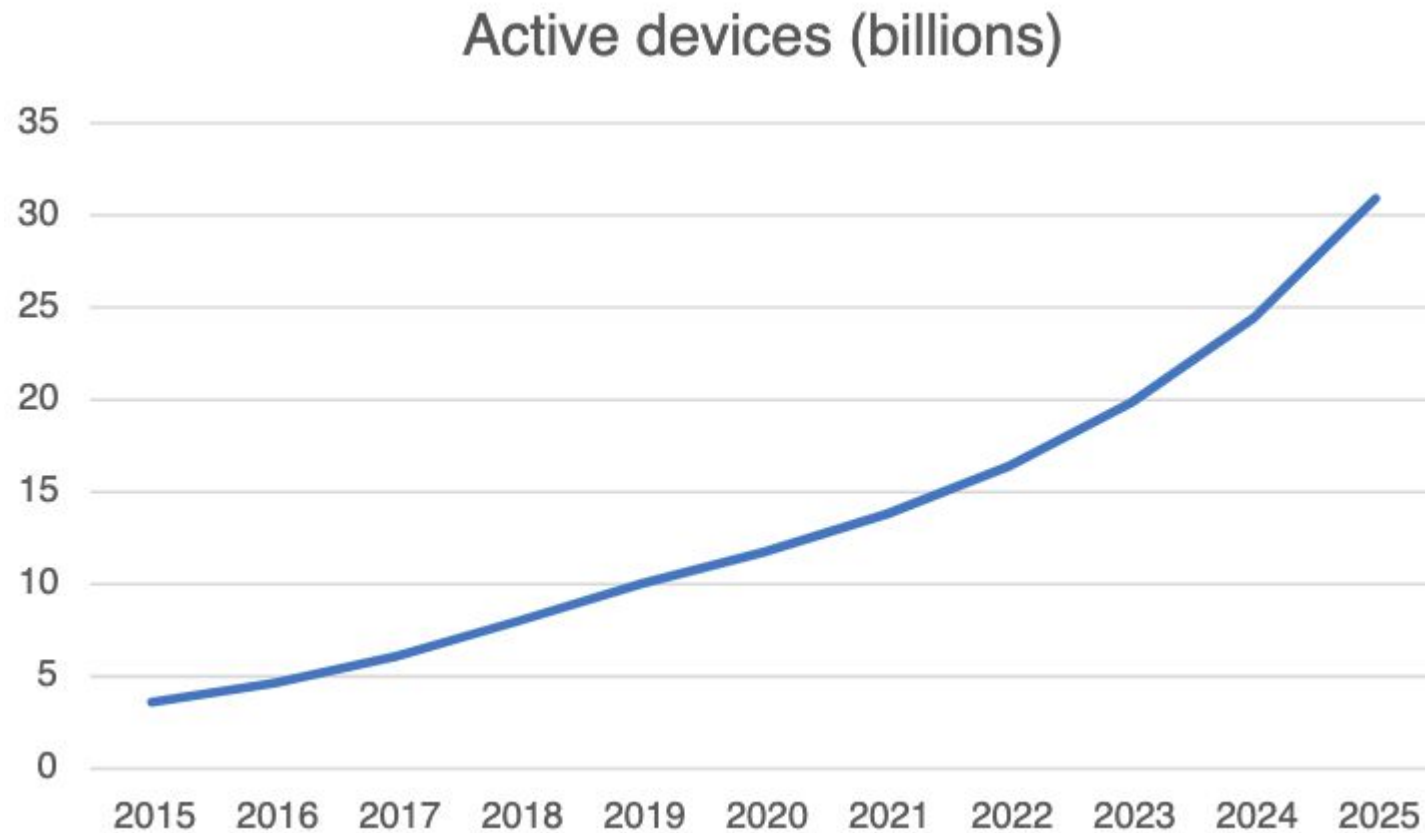


Selection of 40+ IoT Platform providers



Source: IoT Analytics Research 2021; Note: IoT Analytics' definition of an IoT Platform has shifted slightly over time. Condition for republishing: Source citation with link to original post and company website; Non-commercial purposes only

The future



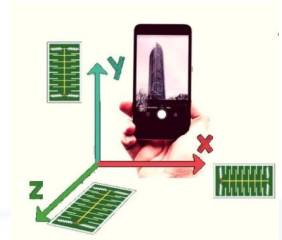
Sensing in IoT

IoT devices

- Sensors (and actuators) can be embedded in devices with limited computing and communication capabilities
- Sensors can be embedded in personal smart objects (smartphones, smartwatches, smartglasses, ...)
- Such devices can be interconnected in sensor networks (IoT)

Physical sensors (some examples)

- **Inertial sensors.** They measure acceleration forces and rotational forces along three axes. Examples: accelerometers, gravity sensors, gyroscopes, and rotational vector sensors.
- **Environmental sensors.** They measure environmental parameters, such as ambient air temperature and pressure, illumination, and humidity. Examples: barometers, photometers, and thermometers.
- **Position sensors.** They measure the physical position of a device. This category includes orientation sensors and magnetometers.

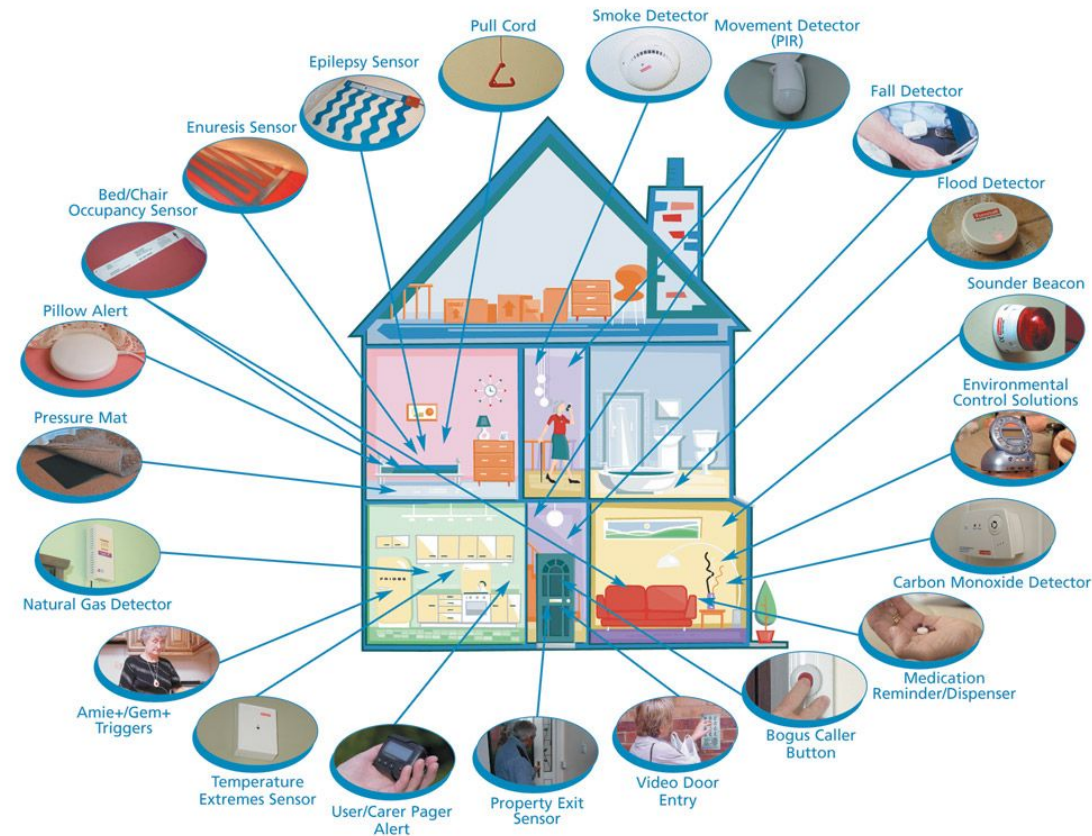


Devices with environmental sensors

- Light, temperature, humidity, magnetic fields, pollution, pressure, electrical fields, sound, chemical elements...



Smart-home sensor devices



<https://csa-iot.org/all-solutions/matter/>

Cameras are also IoT devices

- Many IoT applications may benefit from cameras, like
 - surveillance
 - smart cities (e.g., traffic monitoring)
 - activity recognition
- These devices may be associated with relevant privacy problems in Ambient Intelligence
- In this course, we will only consider sensor-based settings



Mobile/wearable devices

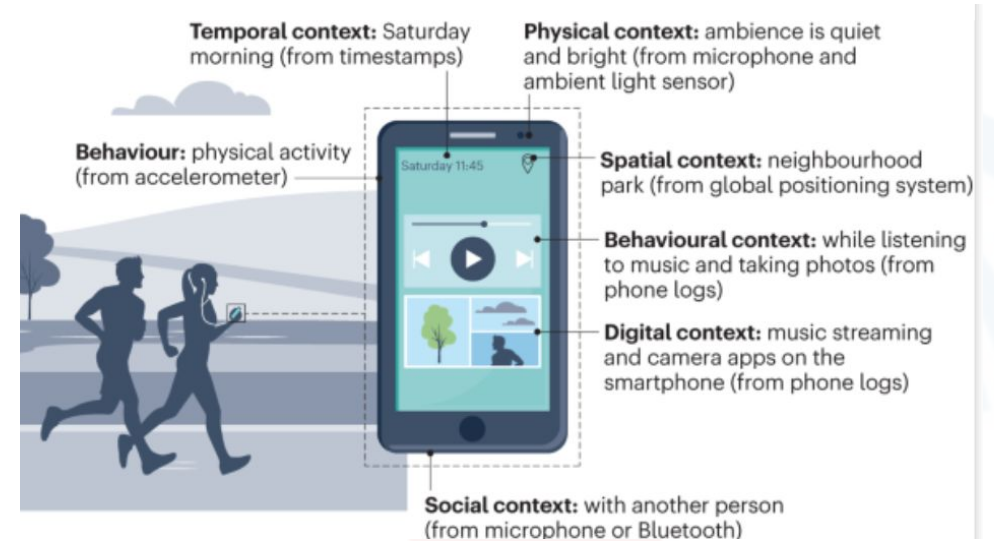
Mobile/Wearable devices have a role in Ambient Intelligence

- Mobile/wearable are also IoT devices providing:
 - sensing
 - gateway and opportunistic data exchange functionalities
 - a possible interface
 - a way to identify the subjects in a smart environment
 - a way to know their goals and preferences



Mobile Sensing

- Mobile devices include several built-in sensors:
 - Inertial sensors (e.g., accelerometer, gyroscope, magnetometer)
 - Environmental sensors (e.g., barometer, light)
 - Location sensors (e.g., GPS)
 - Virtual sensors
- Thanks to communication capabilities, mobile sensor data can be easily integrated with other IoT data



Virtual Sensors

- Services and apps that provide high-level context data from low-level context
- For example, given GPS coordinates there are public web services for:
 - weather (e.g., OpenWeather)
 - semantic location (e.g., Google Places)
 - nearby public transportation (e.g., Transitland)



Wearable Devices

Smart:

- Clothes
- Watches
- Bands
- Clips
- Glasses
- Pendants
- Collars!



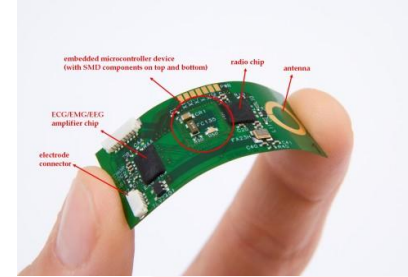
Activity Tracking with wearables

Inertial sensors and GPS data can be used to track the physical activity of the users for well-being applications



Wearable biosensors

- ECG, body temp., blood pressure, heart rate
- Oxygen saturation, perspiration, skin conductivity
- Blood glucose, EMG, EEG...
- More and more integrated in everyday objects



Challenges

- limited resources (energy, CPU, memory, ...)
- different types of interfaces
- high variance in connectivity
- variable location

Sensor Data Management

The importance of Sensor Data Management

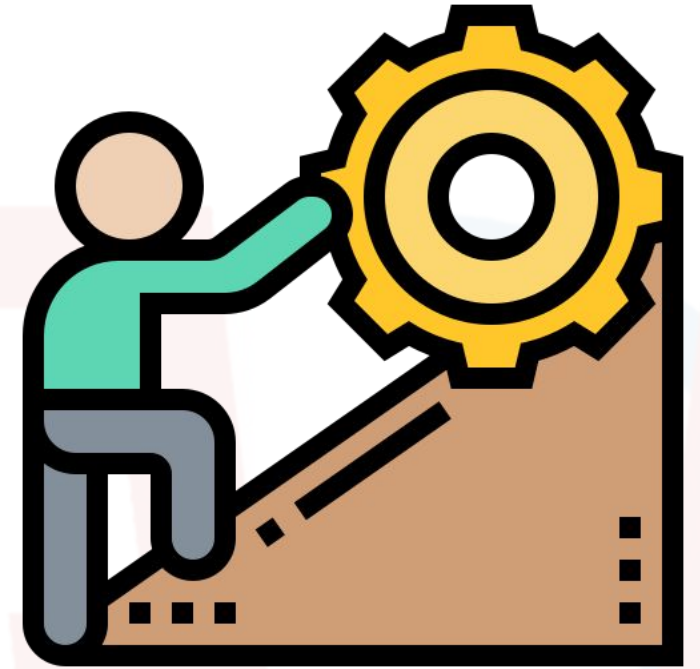
- Intelligent analysis of sensor data is important but...
 - ...collection also plays a challenging role!
 - collecting sensor data is costly from the energy point of view
 - sensor data may be incomplete or may include errors
 - some sensors may sometimes be unavailable
 - sensor data often requires real-time processing
 - need for distributed processing methods

It is crucial to carefully design sensor data management strategies!

Energy Consumption

Sensors are often battery-powered and/or located in inaccessible locations. The following aspects highly impact energy consumption:

- **Data Generation Costs:** Obtaining values from sensors requires high amount of energy (the cost depends on the sensed phenomena).
- **Communication Costs:** the most energy-intensive task is to communicate sensed values to the gateway
 - transmitting 1 bit has the same power cost of 1000 CPU operations



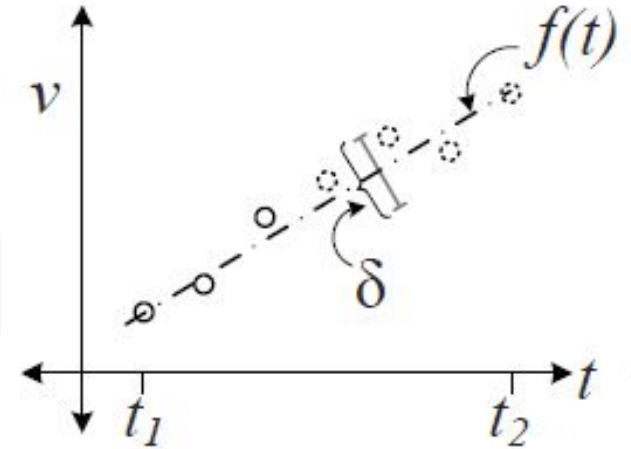
Energy consumption should be reduced as possible!

Sensor Data Management Tasks

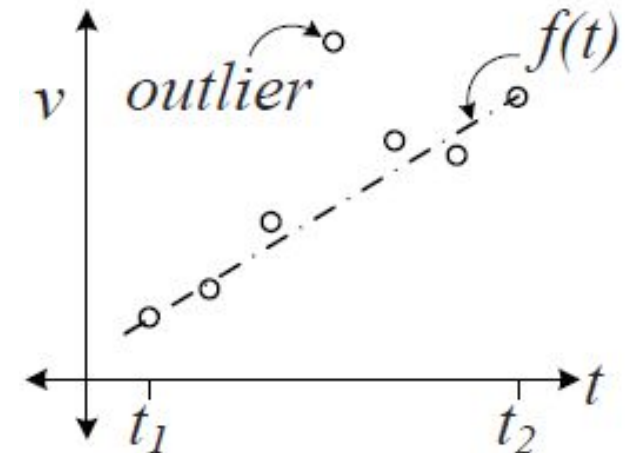
- **Data Acquisition:** how to efficiently acquire sensors samples from a sensor network.
- **Data Cleaning:** how to remove erroneous sensor values due to several potential problems (e.g., sensor failures, loss of communication).
- **Query Processing:** how to obtain desired information, possibly accessing/generating minimal amount of data.
- **Data Compression:** how to summarize the large amount of data generated by sensors.

Tasks

- Data acquisition
 - Goal: efficiently acquiring samples from the sensors
 - Performed locally on the sensor network
- Data cleaning
 - Goal: identifying noisy readings
 - Performed after data acquisition (e.g., base-station)

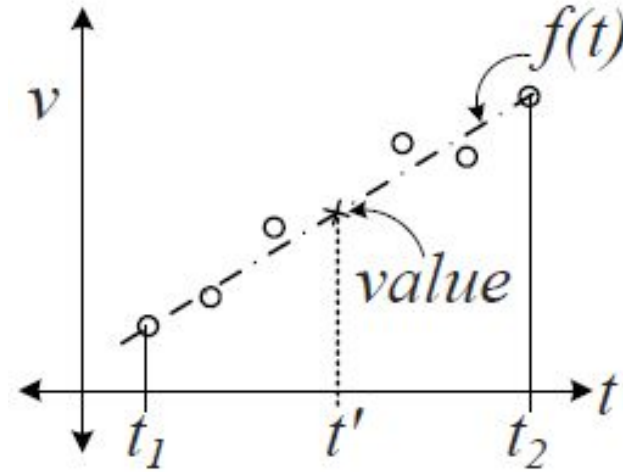


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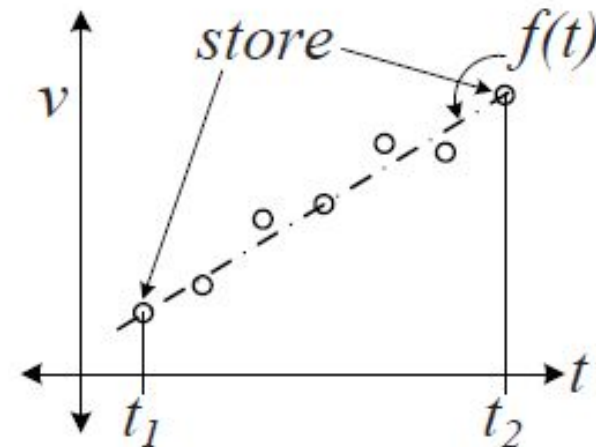


Taks (cont.)

- Query processing
 - Goal: processing queries by accessing or generating minimal amount of data
- Data compression
 - Goal: eliminating redundancy



From [1]



Sensor data acquisition

- Different problem w.r.t. traditional data
 - Need to process continuous stream of sensor data
 - Strong spatio-temporal characterization of data
- Duty cycling may be necessary:
 - Transmitting is costly
 - The device antenna should be put on sleep mode most of the time
 - Coordinated sleep/wakeup scheduling algorithm are required
- Mobility-based:
 - The gateway may be mobile
 - Nodes may be carried by people, cars, animals...
 - Exploit mobility (when available) to better distribute the transmission costs between nodes

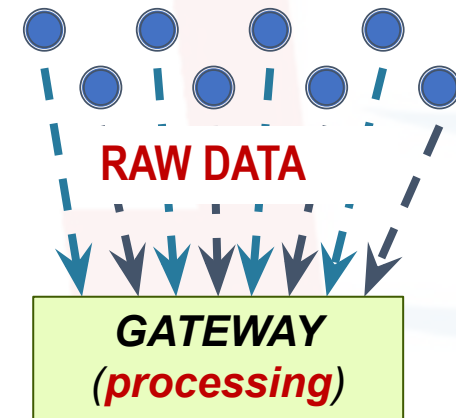
A naive approach for data acquisition: Batch Processing

Buffering on the sensor, and offline processing on the gateway (or remote server)

Precise answer; no processing on the sensor; but high communication costs

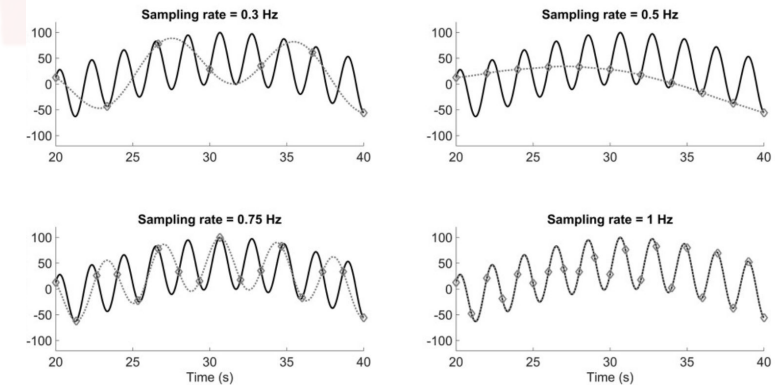
The gateway receives all the data (but with a delay)

SENSORS
(*buffering & communication*)



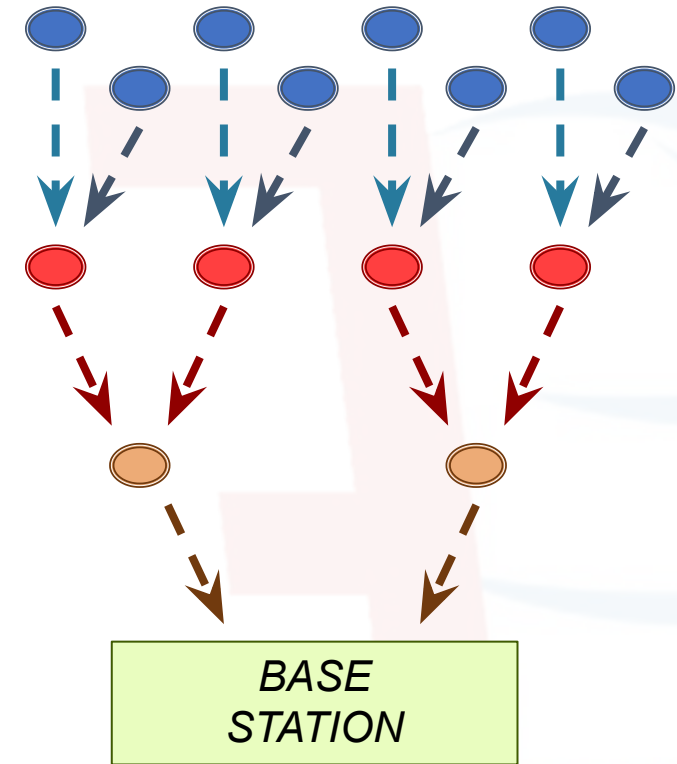
Sampling

- Sampling
 - Not all the sensed signal is used (e.g., read data only at a given frequency)
 - Sensor readings are sampled making sure that considered values are representative for the underlying phenomena
 - e.g., in Human Activity Recognition accelerometers may produce data at 200Hz, but significantly less information (e.g., 50Hz) is sometimes enough



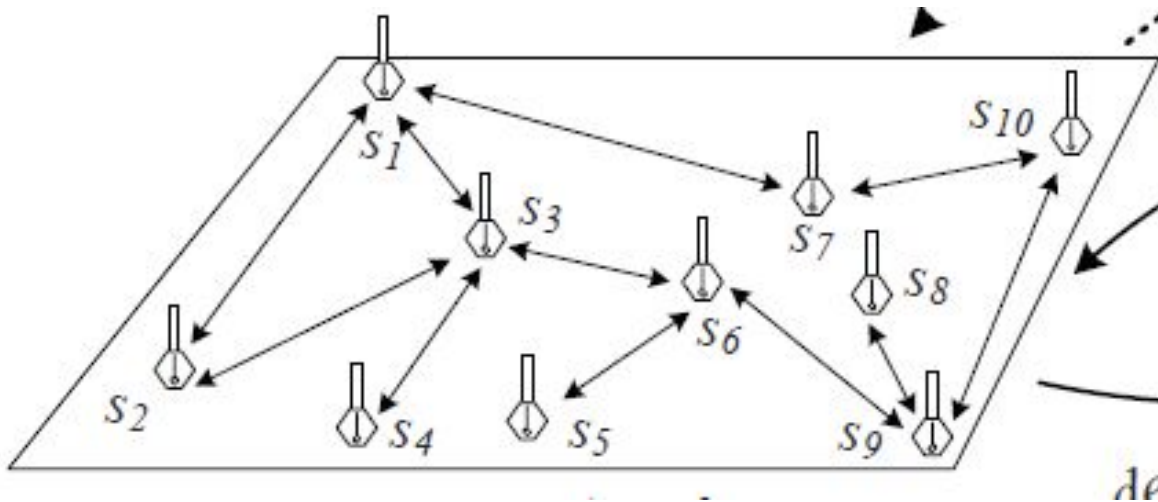
In-Network Query Processing

- In-network query processing
 - (opposed to centralized processing)
 - builds an overlay network
 - performs data aggregation in intermediate nodes to reduce transmission
 - E.g., computing average, min/max, ...
- Objective: significantly reducing the amount of transmitted data over the network!



Pull-based data acquisition

- The user defines the interval and frequency of data acquisition



i	t_i	s_j	x_j	y_j	v_{ij}
1	01:00	1	3.4	7.2	0.1
1	01:00	2	5.2	8.5	0.8
1	01:00	3	7.1	2.2	0.2
2	01:05	1	3.4	7.2	0.7
2	01:05	2	5.2	8.5	0.9
2	01:05	3	7.1	2.2	1.0

sensor_values

From [1]

```
SELECT  $s_j, v_{ij}$  FROM sensor_values SAMPLE INTERVAL 1s FOR 10s
```

Push-based data acquisition

- The gateway and the sensors agree on an expected behavior of the sensor values (i.e., a "model")
 - e.g., the usual temperature fluctuation during the day
- If sensed values deviate from the model, the sensors communicate only the unexpected values

Push-based data acquisition

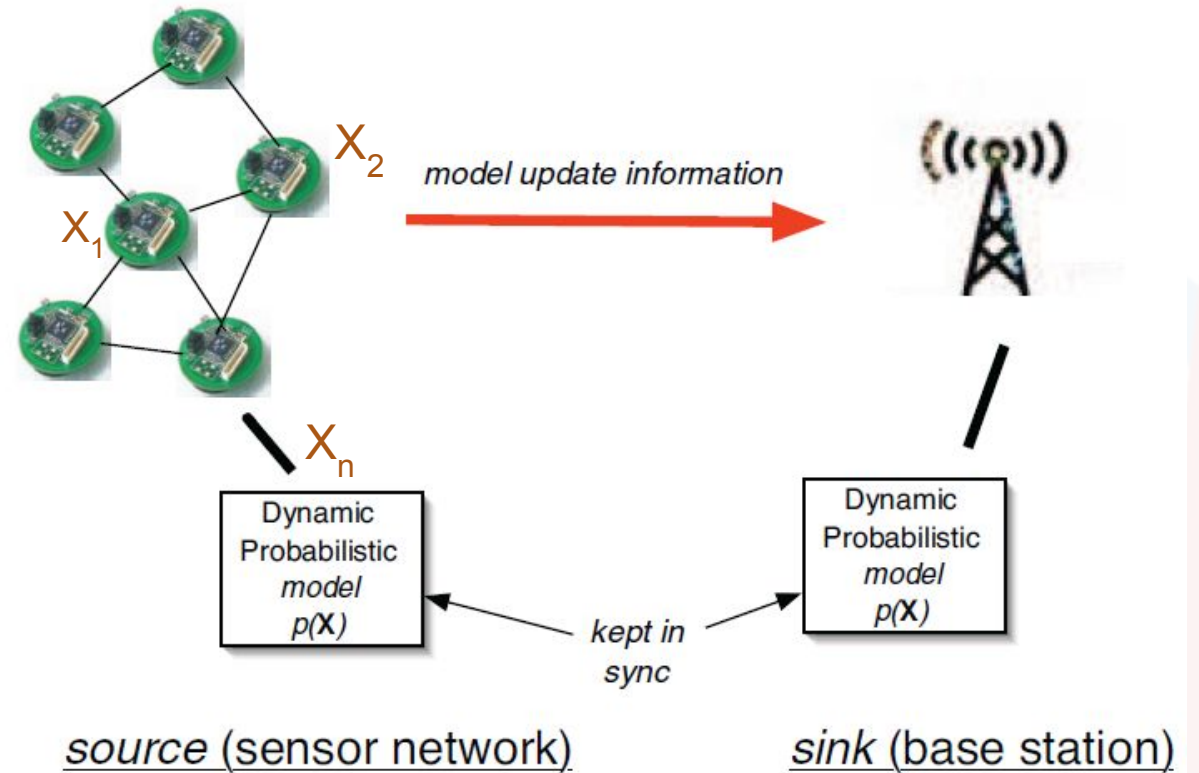
- *Probabilistic models*

- Linear prediction model:

- $\hat{X}_i^{t+1} = \alpha_i \hat{X}_i^t + \beta_i$

- Considers only temporal features

- Treats each sensor independently



From [3]

Push-based data acquisition

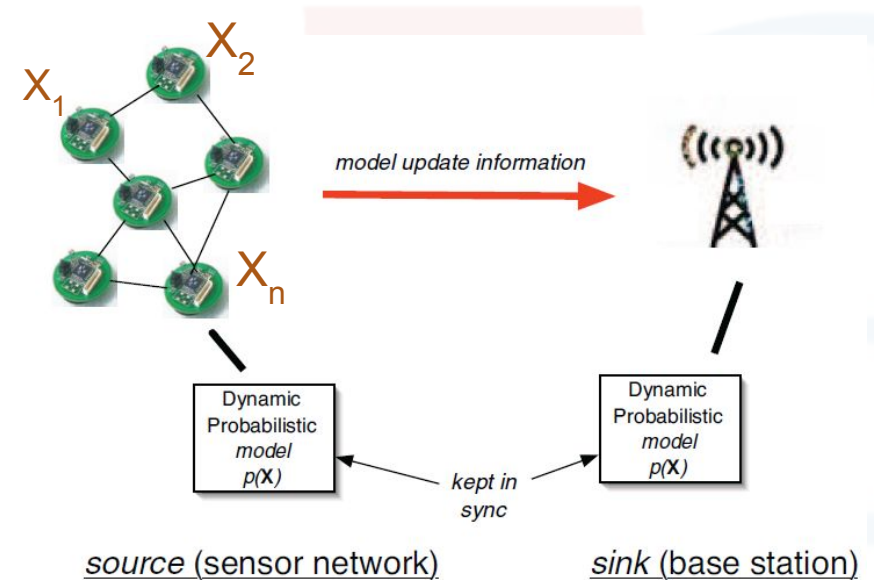
- Markovian models
 - Consider complex (also spatial) dependencies among sensors

- Initial state:

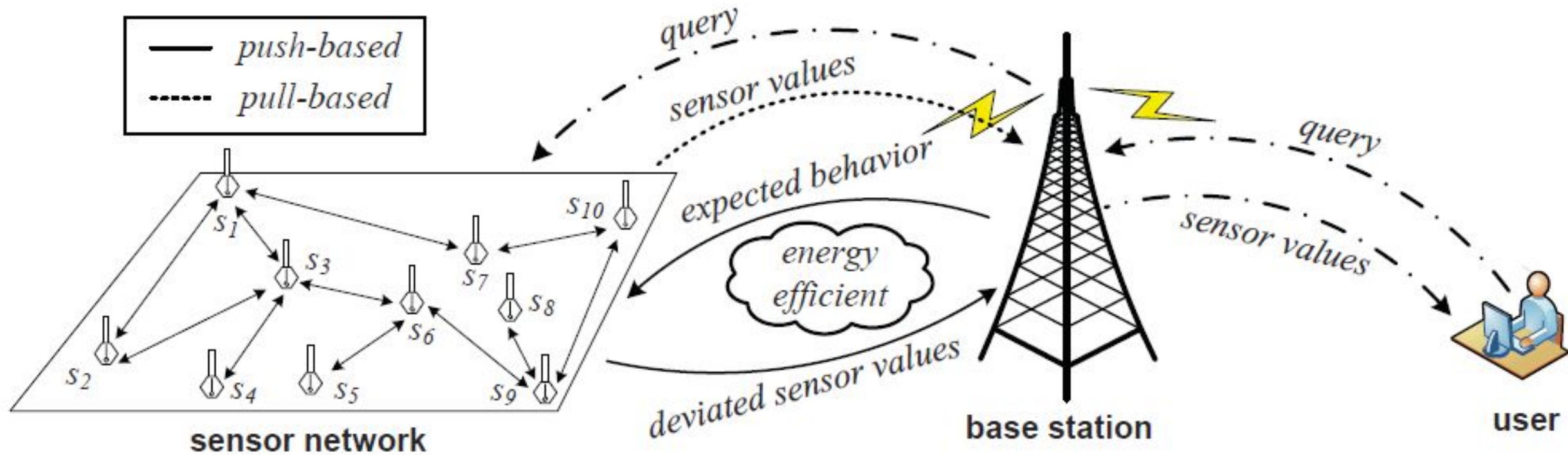
$$p(X_1^{t=0}, \dots, X_n^{t=0})$$

- Transition probability:

$$p(X_1^{t+1}, \dots, X_n^{t+1} \mid X_1^t, \dots, X_n^t)$$



Data Acquisition: Push vs Pull



From [1]

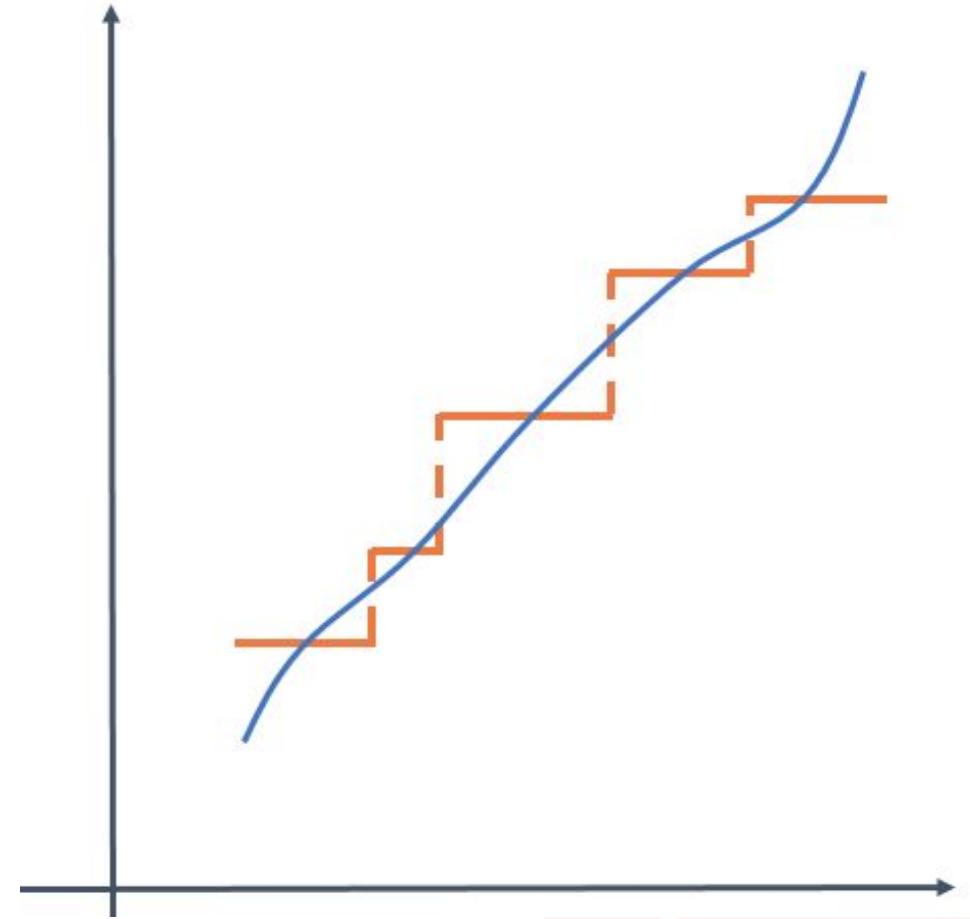
Compression

- Sensors may produce high volumes of data
- Goal: approximate a sensor data stream by a set of functions
- Regression, transformation and filtering can be used to approximate sensor data streams exploiting spatio-temporal correlations
- Orthogonal transformation methods (Fourier or wavelet transform) reduce the data dimensionality



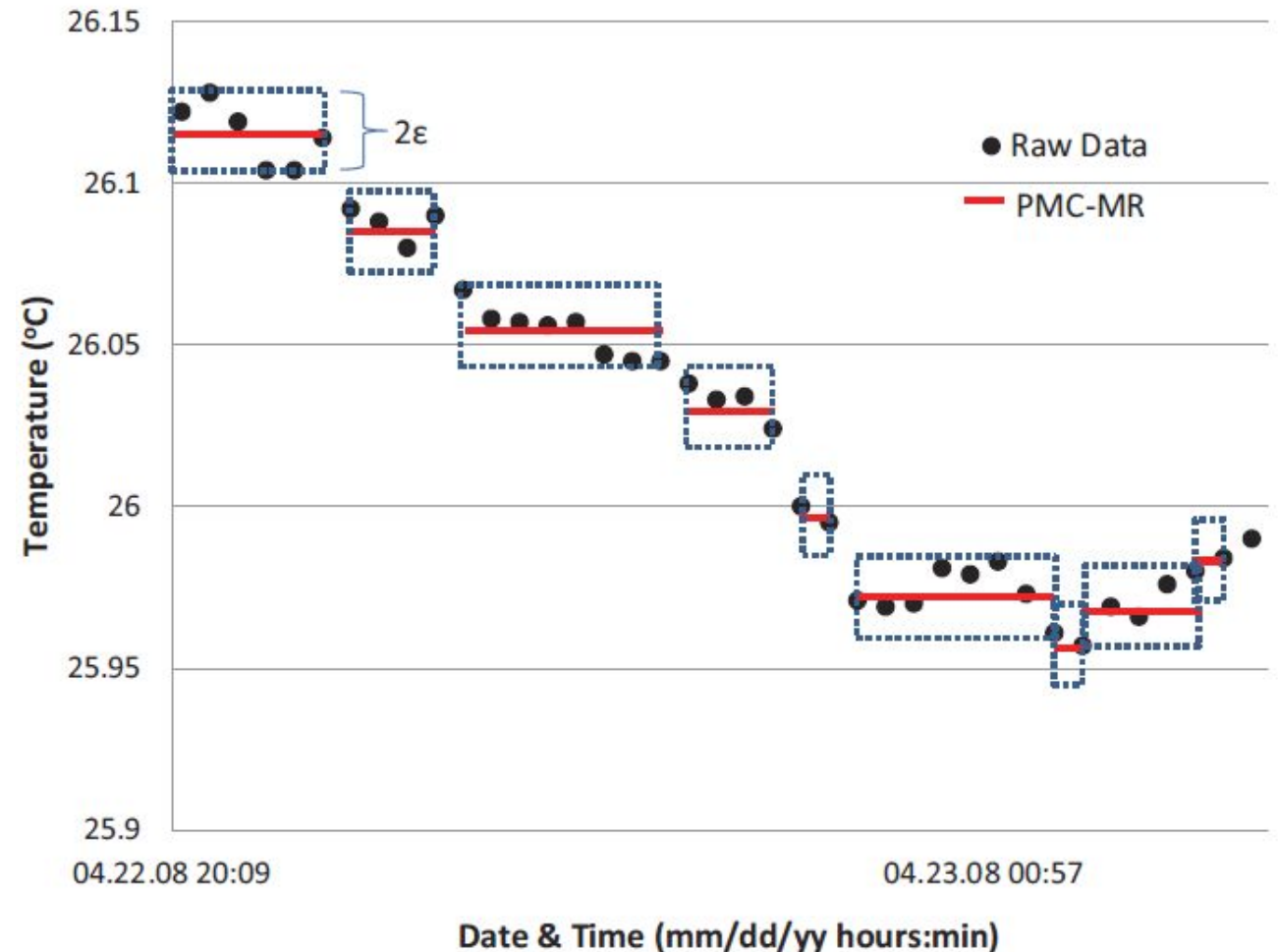
Piecewise Constant Approximation

- Approximates a segment of sensor data with a constant value
 - e.g., first value of the segment (cache filter), mean, median
- The Bounded error (ϵ) guides segmentation:
 - Two consecutive samples $v1$ and $v2$ are in the same segment if $|v1-v2| < \epsilon$



Poor Man's Compression - MidRange (PMC-MR)

- Approximates a data segment with a constant value ((Max-Min)/2 or mean)
- Also in this case, the Bounded error (ϵ) guides segmentation
- Data samples in the same segment should satisfy the following condition: $\max(s) - \min(s) \leq 2\epsilon$



From [1]

Sliding Windows

- Sliding windows
 - Approximate answer based on a group of consecutive readings (e.g., using fixed temporal windows)
 - Run-time computation on the device
 - The gateway receives a continuous stream of approximated sensor readings (with a small delay)

Sliding windows

Example from an accelerometer:

T=0.0	T=0.25	T=0.5	T=0.75	T=1.0	T=1.25	T=1.5	...
Acc-X	Acc-X	Acc-X	Acc-X	Acc-X	Acc-X	Acc-X	...
Acc-Y	Acc-Y	Acc-Y	Acc-Y	Acc-Y	Acc-Y	Acc-Y	...
Acc-Z	Acc-Z	Acc-Z	Acc-Z	Acc-Z	Acc-Z	Acc-Z	...



Features at first second:

- Max/Min/Avg Acc-X,Y,Z, Total-Acc
- Skewness, Kurtosis, Fourier coefficients, ...

Approach used in
accelerometer-based
activity recognition

Overlapping sliding windows

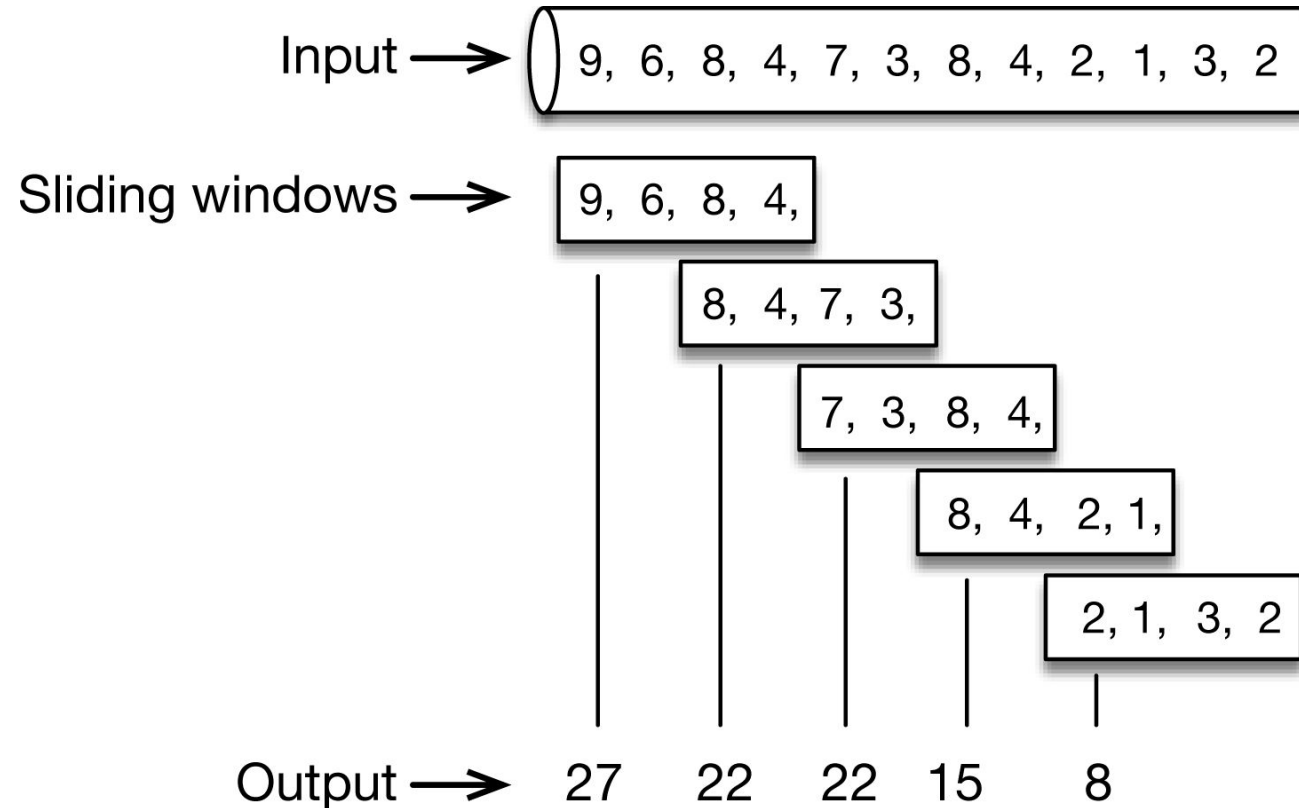
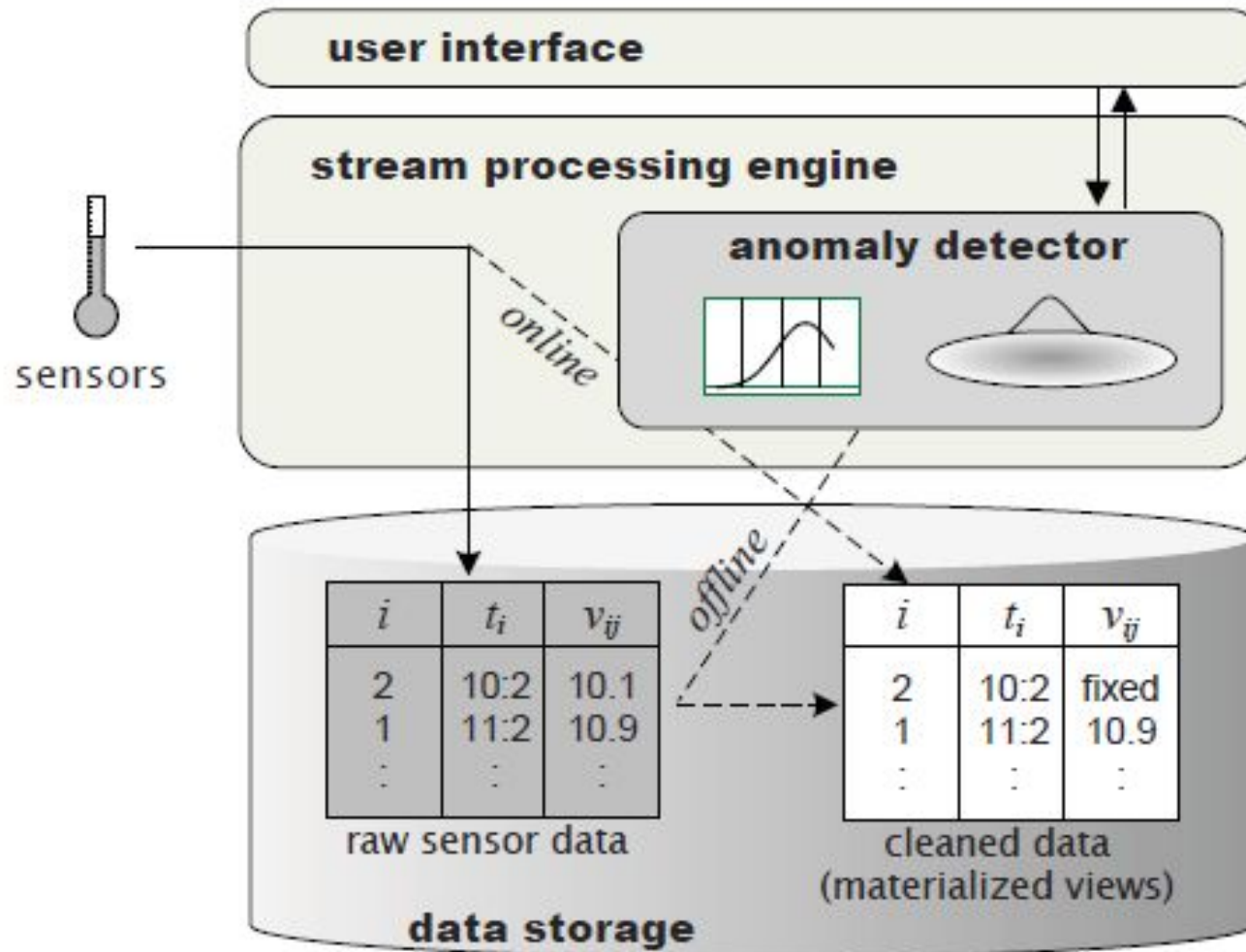


Figure from: Introduction to Apache Flink by Kostas Tzoumas, Ellen Friedman, O'Reilly, 2016

Sensor data cleaning

- Sensor data are uncertain and erroneous
 - discharged batteries, network failures
 - low-quality sensors, freezing/heating of the casing or measurement device
 - accumulation of dirt, mechanical failure or vandalism
- Intelligent data analysis requires cleaning sensor data!
- Note: data cleaning usually occurs ***after data acquisition***
 - it is also always a good practice keeping raw (not cleaned) data in case of mistakes or required changes in the data cleaning process

Data cleaning: a typical pipeline



References

- [1] S. Sathe, T. G. Papaioannou, H. Jeung, K. Aberer: A Survey of Model-based Sensor Data Acquisition and Management. Chapter 2 of: Managing and Mining Sensor Data, Springer 2013, ISBN 978-1-4614-6308-5.
- [2] G. Anastasi, M. Conti, M. Di Francesco, A. Passarella: Energy conservation in wireless sensor networks: A survey. Ad Hoc Networks 7(3): 537-568 (2009) (sections 1, 2, 3, 5)
- [3] D. Chu, A. Deshpande, J. M. Hellerstein, W. Hong: Approximate Data Collection in Sensor Networks using Probabilistic Models. ICDE 2006 (sections 1, 2, 3.1)

